

Multiple Choice (10 marks)

1. The net ionic equation expected when solutions of NH_4Br and AgNO_3 are mixed together is
 - (a) $\text{Br}^-(\text{aq}) + \text{NO}_3^-(\text{aq}) \rightarrow \text{NO}_3\text{Br}(\text{aq})$
 - (b) $\text{NH}_4^+(\text{aq}) + \text{NO}_3^-(\text{aq}) \rightarrow \text{NH}_4\text{NO}_3(\text{aq})$
 - (c) $\text{NH}_4^+(\text{aq}) + \text{Br}^-(\text{aq}) \rightarrow \text{NH}_4\text{Br}(\text{aq})$
 - (d) $\text{NH}_4\text{Br}(\text{aq}) + \text{NO}_3^-(\text{aq}) \rightarrow \text{NH}_4\text{NO}_3(\text{aq}) + \text{Br}^-(\text{aq})$
 - (e) $\text{Br}^-(\text{aq}) + \text{Ag}^+(\text{aq}) \rightarrow \text{AgBr}(\text{s})$
2. At 1000°K , $K = 2.37 \times 10^{-14}$ for the reaction, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$. If you inject one mole of each N_2 and H_2 in a one-liter box...
 - (a) 10.2%
 - (b) 6.5%
 - (c) 8.4%
 - (d) 3.2%
 - (e) 5.0%
3. 40 g of ice at 0°C is mixed with 100 g of water at 60°C , What is the final temperature after equilibrium has been established? Heat of fusion of $\text{H}_2\text{O} = 80\text{ cal/g}$...
 - (a) 40°C
 - (b) 30°C
 - (c) 20°C
 - (d) 50°C
4. What is the conjugate acid of the H_2PO_4^- ion?
 - (a) PO_4^{3-}
 - (b) H_3PO_4
 - (c) H_3PO_3
 - (d) H_4PO_4
 - (e) H_2PO_3^-
5. Suppose 100 ml of oxygen were collected over water in the laboratory at a pressure of 700 torr and a temperature of 20°C . What would the volume of the dry...
 - (a) 70 ml

- (b) 80 ml
- (c) 90 ml
- (d) 83.7 ml
- (e) 76.5 ml

True / False (10 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Determine the molarity of a 40.0% solution of HCl which has a density of 1.20 g/ml.' is '9.95 M'.
2. True or False: The correct answer to 'At standard conditions, 1 liter of oxygen gas weighs almost 1.44 g, whereas 1 liter of hydrogen weighs only .09 g. Which gas diffuses...' is 'Hydrogen diffuses 4 times faster than oxygen'.
3. True or False: The correct answer to 'Calculate spin density on the central carbon atom of malonic acid radical ($\bullet\text{CH}(\text{COOH})_2$) if the hyperfine value for the α -hydrogen atom is...' is '0.35'.
4. True or False: The correct answer to 'Suppose that 2.5 mmol N_2 (g) occupies 42 cm^3 at 300 K and expands isothermally to 600 cm^3 ...' is ' -15 J '.
5. True or False: The correct answer to 'Which of the following molecules is a strong electrolyte when dissolved in water?' is ' CH_3COOH '.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Electron diffraction makes use of electrons with wavelengths comparable to bond lengths. To what speed must an electron be accelerated for it to have a wavelength of 100 pm ?
2. For the general reaction $\text{A} + \text{B} \rightarrow \text{C} + \text{D}$, what is the effect on the number of collisions between A and B of (a) tripling the concentration of each, (b) quadrupling the concentration of each?

End of Paper